

IFYMB004 2023/24

NCUK
THE UNIVERSITY CONSORTIUM

MATHEMATICS (BUSINESS)

INTERNATIONAL FOUNDATION YEAR (IFY)



TABLE OF CONTENTS

Syllabus Overview	2
Assessment.....	3
Assessment Objectives	4
General Learning Outcomes	4
Teaching, Learning and Assessment Guidance	5
Resources	11
Module Content	13
Appendix A - Exemplar Teaching Plan.....	23
Appendix B – Notation List	34
Appendix C - Formulae Booklet	36

SYLLABUS OVERVIEW

Module Code	IFYMB004
Module Name	Mathematics Business
Programme Name	International Foundation Year
Percentage breakdown of Coursework	0%
Percentage breakdown of Exam/Test	100%
Delivery period	The syllabus will usually be delivered over two 15-week semesters
Recommended minimum teaching hours	120 (4 hours per week for 30 weeks)
Recommended minimum independent study hours	120 (4 hours per week for 30 weeks)

AIMS

To develop student's key knowledge, understanding, skills and application of mathematics appropriate for entry to a degree course at an NCUK University.

General Aims

- ⇒ To develop abilities to think logically, to recognise incorrect reasoning and to express ideas clearly.
- ⇒ To develop an enthusiasm for the subject and the skills required to apply the knowledge to both the further study and application of mathematics.
- ⇒ To develop in students an understanding of how theory and application work together.
- ⇒ To develop students' skills in modelling and the interpretation of results.
- ⇒ To develop the necessary English mathematics vocabulary and terminology to use their mathematics knowledge effectively in an NCUK University.
- ⇒ To encourage students towards a level of independence in both the planning and organisation of their studies.
- ⇒ To assist the development of competence and confidence of the students as learners, taking responsibility for their own learning through directed reading and study.

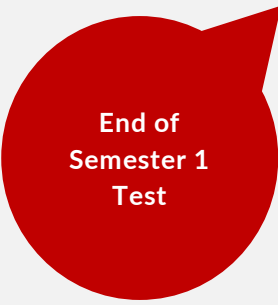
Specific Aims

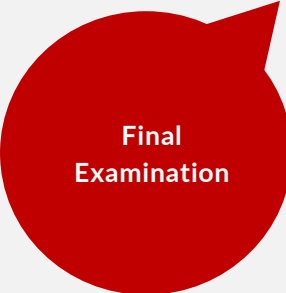
- ⇒ To revise basic skills and develop further skills in algebra.
- ⇒ To demonstrate basic skills in trigonometry and coordinate geometry.
- ⇒ To differentiate and integrate, including the selection and use of appropriate rules and techniques, and the application of the calculus.
- ⇒ To develop concepts in probability and statistics relevant to business planning.
- ⇒ To be confident and competent with the operations of a scientific calculator and its use.
- ⇒ To apply mathematical techniques to simple "real life" problems.
- ⇒ To be familiar with, and competent in, the use of technology to solve pure and applied mathematical problems.

TOPICS OF STUDY

- ⇒ Linear Equations
- ⇒ Quadratic Equations, Remainder Theorem, Equation of a Circle and Inequalities
- ⇒ Binomial Expansions, Sequences and Series
- ⇒ Calculus – Differentiation & Further Differentiation
- ⇒ Introduction to Statistics
- ⇒ Correlation, Linear Regression and Time Series
- ⇒ Financial Mathematics
- ⇒ Simple Probability
- ⇒ Indices, Exponential and Logarithmic Functions
- ⇒ Trigonometric Functions
- ⇒ Calculus – Integration & Further Integration
- ⇒ Further Probability and Set Theory
- ⇒ Probability Distributions

ASSESSMENT

	What is Assessed?	Topics A-H
	Duration/Word Count	2 hours 10 minutes
	Total Marks	100
	Rubric	Approximately 13 compulsory questions of varying length.
	Contribution to Overall Grade	30%

	What is Assessed?	Topics A-O
	Duration/Word Count	2 hours 40 minutes
	Total Marks	125
	Rubric	Approximately 15 compulsory questions of varying length.
	Contribution to Overall Grade	70%

ASSESSMENT OBJECTIVES

	Description	End of Semester 1 Test (30%)	Exam (70%)	Overall
AO1	Knowledge & Understanding: Knowledge of mathematical techniques.	50-60%	50-60%	≈55%
AO2	Application and Communication: Recognising the appropriate procedure and applying combinations of skills and techniques.	40-50%	40-50%	≈45%

GENERAL LEARNING OUTCOMES

On successful completion of this module, a student will be able to:

Knowledge and understanding	⇒ Recognise, recall and apply specific mathematical facts, principles and techniques. ⇒ Select, organise and present relevant information clearly and logically. ⇒ Select and apply appropriate mathematical techniques to solving problems.
Intellectual skills	⇒ Apply mathematical techniques to problems from a variety of relevant discipline areas. ⇒ Present and interpret data in tables, diagrams and graphs. ⇒ Carry out calculations using a formula booklet and calculator. ⇒ Give answers to an appropriate degree of accuracy (i.e. to a specified number of significant figures or decimal places). ⇒ Discuss and interpret results obtained, including an estimate of accuracy.
Practical skills	⇒ Specify what data are required for a given task. ⇒ Collect relevant data in an effective and efficient way.
Transferable skills	⇒ Study independently and make personal notes for problem-solving and revision purposes. ⇒ Source and retrieve information from a variety of locations such as textbooks, websites etc. ⇒ Select and employ problem-solving skills (description, formulation, solution/analysis, interpretation). ⇒ Manage and present data in a variety of formats.

TEACHING, LEARNING AND ASSESSMENT GUIDANCE

Teachers should use a range of different learning and delivery styles in order to give students experience of the types of approach they will encounter in an NCUK University e.g. lectures, tutorials, seminars, laboratory practicals etc. Appendix A (Exemplar Teaching Plan) suggests a delivery format designed to facilitate teaching and learning in each topic. This is provided as a guide and teachers are at liberty to diverge from this plan. The Study Centre may also choose to timetable more than the minimum number of teaching hours (4 hours per week) to meet the needs and abilities of students.

Teachers should take every opportunity to use case study and data response or practice material to develop the ability of students to apply their knowledge and develop their skills of handling and evaluating information. Tutorials should involve both group and individual activities, with a strong emphasis on applying knowledge from lectures and independent study to problem-solving. It is important for all students to have opportunities to speak in English during each tutorial. Suggested activities include students being encouraged to explain in English their answers in pairs or small groups, students providing answers to the whole class whilst standing at the front of the group, student-led presentations, and group activities that require discussion and debate.

Students will have different backgrounds in the subject and it will be necessary to provide opportunities for directed independent study, so as to allow each student to develop at their own pace to reach the required level for the assessments. Activities (homework) for independent study should be set weekly. It is advisable to also consult with EAP teachers when planning homework as there may be subject tasks which can support the development of academic English (and vice-versa).

DEVELOPING INDEPENDENT LEARNING

It is important that the learners are given the guidance and support to develop from pupils within a teacher-led learning environment, to students who feel confident taking ownership of their learning through independent study. As students may have an educational background where concepts of teaching and learning are different to those prevalent in NCUK Universities, aspects of independent learning should be introduced gradually throughout the programme.

Below are suggested examples of ideas which could be used to encourage independent learning:

Acknowledging Failure and Success

A fear of failure or making mistakes can be a significant obstacle which prevents learners from being independent. It is therefore important to acknowledge the benefits of failure with regard to learning early in the module delivery. This could be explored in more depth through practical examples and calculations in which failure leads to success. Teachers should ensure that students are also praised, particularly through formative feedback, and encourage learners to demonstrate development from their past mistakes.

Classroom Reversal

Where possible, teachers might entrust the teaching, practice and revision of sub-topics or items from the syllabus to students themselves. Sharing the responsibility of teaching with students can be a more motivating way to encourage them to take their first steps into independent study than being asked to study independently for purely personal reasons. It also provides a strong reason why independent learning

should be carried out (in this case, to support their peers) and a visible outcome of their independent work (the performance of their peers).

A demonstration of possible student-led tasks and activities can be a useful way to introduce this concept within the classroom. A clear schedule for student-led sessions can also help to make it routine. Sessions could be led by groups initially before asking students to lead them individually when they have grown in confidence.

PAST ASSESSMENTS

Many learners can find it reassuring to revisit past assessments to support their preparation for formative and summative assessments. A selection of past Coursework, End of Semester 1 Test and Final Examination papers are available for this purpose via the NCUK Teachers' Hub. It is important that students are guided to sections and questions which will support the meeting of current learning outcomes or that past assessments are appropriately adapted to ensure their fit to the current syllabus.

While classroom time can be allocated to the review of and use of past papers, students should also have the opportunity to make use of these for independent study. Teachers should work with colleagues to decide which papers may be kept aside for use in formative assessment or class time and which to release to students for independent study.

There are a number of ways past assessments can be used to support learning in addition to recalling knowledge. These include:

- ⇒ An exploration of incorrect or incomplete answers which could highlight gaps in understanding and help to inform future teaching and learning
- ⇒ Identifying whether the correct structure is being used to answer extended answers
- ⇒ Practicing answers to certain questions within particular time constraints
- ⇒ Distinguishing different command words, such as 'explain', 'describe' and 'evaluate'
- ⇒ Allowing students to notice weaknesses in their own knowledge or answers by allowing them to self-mark with the assessment criteria
- ⇒ Allowing students to establish a greater familiarity with the assessment criteria and what they are expected to do to achieve a certain grade

SUPPORTING ENGLISH LANGUAGE DEVELOPMENT

It is important for the development of academic literacies that the delivery of the subject material is integrated with the English for Academic Purposes (EAP) module (or EAP for Proficient Users or Research and Communication Skills modules where appropriate). Regular communication between the subject module teacher(s) and the EAP, EAPPU or RCS teacher(s) will provide a basis on which to support and guide students. Students will benefit from collaborative activities, where the subject module and EAP/EAPPU/RCS teachers jointly deliver classes in relation to activities such as essay or report writing style and using academic sources.

There are also a number of simple tasks and approaches which can be built into this module which can support language development. These include:

Introducing New Vocabulary

When students are introduced to new vocabulary in class, ensure that these words are not introduced in isolation, but in a chunk or phrase within a wider understood context. As an example, if students were given the following extract from the Economist (2021), they may be unfamiliar with the word 'ailments':

The cholera pandemic of the early 1830s hit France hard. It wiped out nearly 3% of Parisians in a month, and hospitals were overwhelmed by patients whose ailments doctors could not explain.

If this word was then to be explored and recorded, it would be better to look at this chunk:

...hospitals were overwhelmed by patients whose **ailments** doctors could not explain.

By giving students a clear context, it is easier to initially discern meaning while also making it more memorable for revision. By keeping the word within a larger chunk, there are also a number of grammatical and lexical relationships which can be established by learners:

- ⇒ Common collocations; explain ailments, a patient's ailments.
- ⇒ The inclusion of 'ailments' in the semantic field of medicine / health.
- ⇒ The determiner 'whose' implies that 'ailments' is a noun and the 's' that it is a plural, countable noun.
- ⇒ Wider examples of past passive and defining relative clauses.

Although it should not be expected that this level of exploration is conducted within a subject module, highlighting new vocabulary in chunks does give language learners a greater opportunity to notice features of language at their own speed. It also provides a grammatical template which students can build on when writing their own answers.

Highlighting Stressed Syllables

One way to support the development of students spoken fluency and overall rhythm is to draw attention the stressed syllable within key, subject-specific vocabulary. This can be done very simply on the board in a number of ways and you may wish to discuss this with the EAP/RCS teacher(s) at your Study Centre in order to align your preferred method(s). As you may be students' main model for how to pronounce subject-specific vocabulary, it is essential they have a clear record which they can refer to during independent study.

Encouraging Functional & Discursive Language

When completing written and spoken tasks, such as group discussions or the presentation of ideas within class, you may wish to encourage students to use functional or discursive language that will help them to structure and explain their ideas with greater coherence and clarity. For instance, this may include language for agreeing /disagreeing, clarifying what they have said when they are unclear, or how to highlight a particular point. This could be achieved by:

- ⇒ Asking students to take out, look at and use language explored for this purpose within their EAP classes. Collaboration with the EAP teacher(s) to decide when in the module particular language will be covered and for what purpose will greatly support students' language development.

- ⇒ Make functional and discursive language relevant to your subject available in the classroom for students to use. Adding useful language to walls or 'vocabulary mats' on desks can remind students of language available to them while also encouraging independence.

Allowing Learners Time to Speak

Although some students may be reluctant to speak in class, it is also important to remember that less proficient speakers require time to construct utterances and may need some time to prepare themselves. While silence in a classroom can sometimes feel awkward, try to avoid filling this silence with teacher talk time, particularly when you are expecting answers from students. You may want to prepare students to speak by giving them advance warning, for example, by giving the class time to make notes before being asked to speak. Alternatively, you may wish to select students to answer particular questions in advance.



FORMATIVE ASSESSMENT

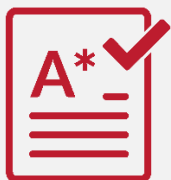
It is important that students are given the opportunity to engage in and submit formative assessments and receive feedback on this work. Formative assessment should be designed to inform students of their progress and enable them to develop and practice coursework and examination skills.

In order to assist teachers in setting formative assessment, NCUK provides the following for this module:

- ⇒ Exemplar assessments – a full set of exemplar assessments with mark schemes (End of Semester 1 Test, Coursework and Final Examination) are downloadable from the Teachers' Hub. These can be used as mock assessments for students throughout the academic year.
- ⇒ Past exam questions – a set of past examination questions is accessible for teachers via the Teachers' Hub. Teachers can generate their own papers for use as practice exams, pick out specific questions to work through with students in class, or set as homework.

In addition to the above, teachers should set regular homework tasks that allow students to understand their strengths and weaknesses and should provide advice to students on how to improve in areas of weakness.

Beyond developing students' skills, the crucial element of formative assessment is the feedback provided by the teacher. In some instances, it may also be appropriate for students to provide feedback on each other's work in a group setting.



SUMMATIVE ASSESSMENT

The breakdown of summative assessment for this module and the rubrics for each component are given on page 3 of this document. All summative assessments are provided by NCUK.

End of Semester 1 Test

The End of Semester 1 Test will cover topics from Semester 1 only and will take place under examination conditions. Teachers at the Study Centre will be responsible for marking the End of Semester 1 Tests in accordance with the NCUK issued mark scheme.

Examination

The examination will cover topics from both semesters. Teachers at the Study Centre will be responsible for marking the Examination in accordance with the NCUK issued mark scheme.

An NCUK approved calculator is permitted in the examination (refer to NCUK policy '*Calculator Regulations*').

It is the Study Centre's responsibility to provide the following materials for the End of Semester 1 test and Examination:

- ⇒ Graph Paper
- ⇒ Formula Booklet (refer to *Appendix C* of this syllabus)

Supporting Documents

It is essential that coursework and examinations are administered in accordance with NCUK regulations. Please refer to the following sections of the NCUK Academic Handbook for details:

- ⇒ Coursework Administration and Regulations
- ⇒ Centre Marking and Recording Results
- ⇒ Academic Misconduct Policy
- ⇒ Examination Administration

RESOURCES



Core Textbooks

CGP Books (2017), *Mathematics for A level The Textbook*, Coordination Group Publications Ltd (CGP)

ISBN: 9781782947233 (Print)

Electronic version available to purchase [here](#)

CGP Books (2018), *AS and A level Further Maths for Edexcel Complete Revision and Practice*, Coordination Group Publications Ltd (CGP)

ISBN: 9781782948698 (Print)

Each copy comes with a free online edition.



Further Reading

An introduction to Mathematics written in the style of a travelogue:

Bellos, A. (2020), *Alex's Adventures in Numberland*. Bloomsbury Publishing

ISBN: 9781526623997 (Print)

ISBN: 9781408811146 (eBook)

Centre for Innovation in Mathematics Teaching A-level Course Material:

<https://www.cimt.org.uk/projects/mepres/alevel/alevel.htm>



Useful Websites

Free mathematics software:

<http://www.geogebra.org/>

<https://www.desmos.com/>

A variety of resources sorted by topic:

<https://www.mathcentre.ac.uk/>

A level maths resources:

<https://www.stem.org.uk/resources/curated-collections/secondary-and-level-mathematics-0>

A level maths revision resources:

<https://app.senecalearning.com/courses?Price=Free&Subject=Maths&Age+Group=A+Level>

For students and teachers to pose or answer any maths problem at any level:

<http://math.stackexchange.com>



Teacher Resources

NCUK Teacher's Hub – accessible via the NCUK learning and assessment platform – please contact your Study Centre Academic Manager for access.

MODULE CONTENT

All topics must be covered. Appendix A provides an Exemplar Teaching Plan which indicates the proportion of time to be spent on each topic and activities to support student learning.

The table below details the topics and sub-topics to be covered. A set of specific learning outcomes (SLOs) are given for each topic.

TOPIC A – LINEAR EQUATIONS		
CONTENT	SLOs	
⇒ The equation of a line, parallel and perpendicular lines. ⇒ Solving pairs of simultaneous equations using elimination, substitution and graphical methods.	A1	Find the equation of a straight-line using co-ordinate geometry.
	A2	Find parallel and perpendicular lines and sketch appropriate graphs.
	A3	Find the distance between two points and the co-ordinates of the midpoint.
	A4	Solve pairs of simultaneous equations using elimination, substitution and graphical methods.

TOPIC B – SIMPLE PROBABILITY		
CONTENT	SLOs	
⇒ Define probability, use sample space diagrams to help calculate probabilities. ⇒ Combining probabilities and using tree diagrams. ⇒ (Knowledge of conditional probability such as $P(A B)$ is not expected in this module.) ⇒ (The use of the symbols $\cap$ and $\cup$ will not be expected in this module)	B1	Find the probability of a single event.
	B2	Recognise that $P(A)$ and $P(A')$ mean the probabilities of event A occurring and event A not occurring respectively.
	B3	Find, for two events A and B , the probabilities of both A and B occurring, and the probabilities of either A or B occurring.
	B4	Construct and use a tree diagram.

TOPIC C – QUADRATIC EQUATIONS, REMAINDER THEOREM, EQUATION OF A CIRCLE AND INEQUALITIES

CONTENT	SLOs	
⇒ Quadratic Functions: factorising, completing the square and using the quadratic formula. ⇒ Remainder Theorem: simple algebraic division; use of the Factor Theorem and the Remainder Theorem. Graphs of quadratic and cubic functions. Geometrical interpretation of algebraic solutions of equations. ⇒ Equation of a Circle: in the form $(x - a)^2 + (y - b)^2 = r^2$. ⇒ Inequalities: manipulating inequalities, solving linear and quadratic equations and inequalities.	C1	Solve quadratic equations using factorising, completing the square and the quadratic formula.
	C2	Carry out the process of completing the square to locate vertices (turning points) of graphs.
	C3	Use the discriminant to determine the number of real roots.
	C4	Use surds to give exact solutions.
	C5	Use algebraic division by a monomial or quadratic function.
	C6	Sketch the graphs of quadratic and cubic functions.
	C7	Use the Factor Theorem. Use the Remainder Theorem to determine the remainder when a polynomial is divided by $(ax + b)$.
	C8	Find the radius and co-ordinates of the centre of a circle given its equation and vice versa.
	C9	Recognise that the radius of a circle is perpendicular to a tangent.
	C10	Solve by substitution a linear and quadratic pair of simultaneous equations and apply this to finding the intersection of a line and a curve/circle.
	C11	Recognise and solve linear/quadratic equalities and inequalities.
	C12	Use algebraic and graphical methods to solve inequalities.
	C13	Recognise and distinguish between open and closed intervals (candidates will be expected to understand the word “integer”).

TOPIC D – INDICES, EXPONENTIAL AND LOGARITHMIC FUNCTIONS		
CONTENT	SLOs	
⇒ Laws of Indices for all rational exponents. ⇒ Exponential function: ○ exponential function and its graph ○ introduction to rates of growth ○ solution of equations involving exponential functions ⇒ Logarithmic function: ○ rules and manipulation of logarithms ○ logarithmic function and its graph ○ relationship between exponential / logarithmic functions ○ solution of equations involving either exponential or logarithmic functions	D1	Use and apply equivalences $x^a \times x^b \equiv x^{a+b}$, $x^a \div x^b \equiv x^{a-b}$, $(x^a)^b \equiv x^{ab}$ $x^{-n} \equiv \frac{1}{x^n}$, $x^{a/b} \equiv \sqrt[b]{x^a}$.
	D2	Express solutions in exact form.
	D3	Use a calculator to evaluate exponential and logarithmic expressions.
	D4	Sketch the graphs of $y = e^x$, $y = e^{-x}$ and $y = a^x$.
	D5	Apply exponential functions to problems.
	D6	Apply the rules of logarithms to problems.
	D7	Change the base of a logarithm.
	D8	Solve equations involving exponential and logarithmic functions including those involving “hidden” quadratics.

TOPIC E – BINOMIAL EXPANSIONS, SEQUENCES AND SERIES		
CONTENT	SLOs	
⇒ Binomial expansions: ○ Pascal's triangle ○ factorials ○ binomial expansion (positive integer powers, binomial coefficient notation, evaluation of specific terms) (Candidates could be asked to write down the first few terms in ascending powers of x of an expansion with $n \gg 5$. The meaning of "ascending" or "descending" will always be given. They will be expected to understand the word "coefficient") ⇒ Sequences and series: ○ sequences ○ series ○ sigma notation ⇒ Finite Arithmetic Progressions (AP) and series including sum. ⇒ Geometric Progressions (GP) and series including sum. ⇒ Convergence and divergence of geometric series.	E1	Expand $(1 + x)^n$ for small positive integer n .
	E2	Use Pascal's triangle to find binomial coefficients.
	E3	Expand $(a + b)^n$ where n is a small positive integer.
	E4	Evaluate specific terms of the expansion of $(a + b)^n$ where n is a positive integer.
	E5	Understand the idea of a sequence of terms using general formulae and recurrence relations.
	E6	Use sigma notation for series representations.
	E7	Recognise and sum a finite arithmetic series.
	E8	Recognise and sum a geometric series.
	E9	Define, explain and test for convergence of a series.
	E10	Use an AP or GP to solve practical problems.

TOPIC F – TRIGONOMETRIC FUNCTIONS		
CONTENT	SLOs	
⇒ Angles (degree/radian measure). ⇒ Trigonometric ratios, trigonometric functions (sine, cosine, tangent) and their graphs. ⇒ Solutions of simple trigonometric equations. ⇒ The identity $\cos^2 x + \sin^2 x \equiv 1$. ⇒ $\tan x = \frac{\sin x}{\cos x}$.	F1	Convert from radians to degrees and vice versa.
	F2	Recognise exact values of the trigonometric functions of 0, 30, 45, 60 and 90 degrees and their radian equivalents.
	F3	Find sin, cos and tan of any angle and sketch their graphs.
	F4	Use the area of a triangle formula $\frac{1}{2}ab \sin C$.
	F5	Calculate inverse trigonometric functions.
	F6	Find particular solutions of simple trigonometric equations. (These equations will take the form: $a \sin bx = c$; $a \cos (x + b) = c$; $a \tan^2 bx = c$ over any range.)
	F7	Recognise that $\cos^2 x + \sin^2 x \equiv 1$ and $\tan x = \frac{\sin x}{\cos x}$ (extensive use of these properties is not required at this stage).
	F8	Apply the sine and cosine rules to an arbitrary triangle.

TOPIC G – CALCULUS - DIFFERENTIATION		
CONTENT	SLOs	
⇒ Gradients of tangents and normals to curves, limit form, polynomial rules (inc. First Principles). ⇒ Derivatives of simple functions (exponential, log, trigonometric. The trigonometric functions are $\sin x$, $\cos x$ and $\tan x$ only). ⇒ Use of Formula Booklet (see Appendix C). ⇒ Using derivatives to help sketch curves. ⇒ Equations of tangents and normals. Maxima, minima and points of inflexion which are stationary points. ⇒ Use of the second derivative.	G1	Evaluate the gradient of a curve at a point.
	G2	Recognise and explain the notation $\frac{dy}{dx}$ and $f'(x)$.
	G3	Sketch the derivative graph $\frac{dy}{dx}$.
	G4	Apply the limit formula to simple functions (first principles). (This will be confined to single powers of x .)
	G5	Differentiate any polynomial and any power of x .
	G6	Use the formula booklet to obtain derivatives of standard functions, including e^{kx} where k is a constant.
	G7	Explain second-derivative notation.
	G8	Apply second derivatives to practical problems.
	G9	Find stationary points for a given function.
	G10	Distinguish between local maximum, local minimum and point of inflexion which are stationary points including application to curve sketching. (Points of inflexion will be identified only in instances of zero gradient.)
	G11	Apply SLOs G8 and G9 to practical optimisation problems.
	G12	Obtain the equation of tangent and normal of a curve at a specified point.

TOPIC H – CALCULUS - INTEGRATION		
CONTENT	SLOs	
⇒ Inverse of differentiation, standard integrals (polynomial, trigonometric, exponential), indefinite and definite integration. (The trigonometric functions are $\sin x$ and $\cos x$ only.) ⇒ Area under a curve.	H1	Identify integration as the inverse of differentiation.
	H2	Use the formula booklet to determine indefinite integrals including e^{ax} and $\frac{1}{ax+b}$ where a and b are constants.
	H3	Form and explain the definite integral.
	H4	Evaluate definite integrals.
	H5	Find the area between a curve and the x -axis, including areas partly above and partly below the axis.
	H6	Find the area between a curve and a straight line and the area between two curves.

TOPIC I – INTRODUCTION TO STATISTICS		
CONTENT	SLOs	
⇒ Presentation of statistical data. ⇒ Histograms and the cumulative frequency polygon and curve. ⇒ Data summaries: mode, median, mean and range. ⇒ Standard deviation. ⇒ Quartiles and interquartile range.	I1	Distinguish between continuous and discrete data.
	I2	Draw a line graph using discrete data, a histogram using continuous data.
	I3	Evaluate mode, median, mean and range.
	I4	Understand what “standard deviation” means.
	I5	Evaluate the standard deviation (divisor n).
	I6	Evaluate the mean and standard deviation of data in a frequency distribution, including grouped frequencies.
	I7	Produce and interpret a cumulative frequency distribution.
	I8	Use a cumulative frequency graph to estimate the median, the quartiles and the interquartile range of a set of data.
	I9	Identify a distribution which appears to be skewed.

TOPIC J – FURTHER PROBABILITY AND SET THEORY

CONTENT	SLOs	
⇒ Further Probability: mutually exclusive events and independent events. ⇒ Laws of Probability. ⇒ Conditional Probability. ⇒ Set Theory: sets, intersections, unions, complements. ⇒ Venn diagrams, including their use to solve probability problems. ⇒ (the understanding of set notation will be expected).	J1	Explain set definitions.
	J2	Compose two sets by union or intersection.
	J3	Illustrate sets by using Venn diagrams.
	J4	Use the laws of probability.
	J5	Identify mutually exclusive and independent events.
	J6	Compute conditional probabilities.
	J7	Illustrate probabilities using Venn diagrams and more complicated tree diagrams.
	J8	Calculate the probabilities of combined events

TOPIC K – CORRELATION, LINEAR REGRESSION AND TIME SERIES

CONTENT	SLOs	
⇒ Correlation: scatter graphs. ⇒ Calculation and interpretation of the coefficient of correlation. ⇒ Linear Regression: calculation of the equation of a least-squares linear regression line. ⇒ Time Series: trend-line, moving averages. ⇒ (Calculating seasonal variations and residuals will not be expected.)	K1	Explain the term “correlation” in relation to data sets where both variables must be random.
	K2	Compute the product moment correlation coefficient and evaluate the result in relation to the appearance of the scatter graph; with reference to values close to -1, 0 and 1.
	K3	Compute the equation of a least-squares linear regression line for random variable Y and non-random variable X .
	K4	Be able to interpret in context the uncertainties of estimating a value.
	K5	Construct time series charts.
	K6	Calculate moving averages of a time series.
	K7	Construct the trend-line of a time series.
	K8	Extrapolate a trend line but be aware of the dangers.

TOPIC L – PROBABILITY DISTRIBUTIONS		
CONTENT	SLOs	
⇒ Discrete random variables: probability distributions given algebraically or in tables. ⇒ Calculate the mean $E(X)$ and the variance $\text{Var}(X)$. ⇒ Distributions: binomial distribution. ⇒ Normal distribution and confidence intervals.	L1	Recognise and describe the nature of a “distribution”.
	L2	Construct a probability distribution relating to a given situation involving a discrete random variable X .
	L3	Evaluate the expected value and variance of a linear function of a random variable.
	L4	Distinguish between discrete and continuous distributions.
	L5	Describe the binomial distribution.
	L6	Perform calculations with the binomial distribution.
	L7	Use the binomial cumulative distribution tables.
	L8	Perform calculations using the Normal distribution.
	L9	Convert general data into the standardised form.
	L10	Set up a confidence interval for a mean where the background distribution of any samples will be Normal with a known standard deviation.

TOPIC M – FINANCIAL MATHEMATICS		
CONTENT	SLOs	
⇒ Percentage and percentage change. ⇒ Interest. ⇒ Appreciation and Depreciation.	M1	Carry out any calculation involving direct percentage, reverse percentage, percentage error and percentage change.
	M2	Perform calculations involving simple and compound interest.
	M3	Calculate appreciation and depreciation using knowledge of a geometric progression gained in E8.
	M4	Estimate when a certain value is reached in an appreciation or depreciation situation using knowledge of logarithms gained in D6.

TOPIC N – FURTHER DIFFERENTIATION		
CONTENT	SLOs	
⇒ Rules: sum, product, quotient rules and the chain rule (composite functions). ⇒ Implicit differentiation.	N1	Apply the product rule, quotient rule and chain rule.
	N2	Find $\frac{dy}{dx}$ of an implicit function.
	N3	Understand the result $\frac{d}{dx} a^x = a^x \ln a$.

TOPIC O – FURTHER INTEGRATION		
CONTENT	SLOs	
⇒ Integration by substitution. Change of variable. Use of Formula Booklet (see Appendix C). ⇒ Partial fractions (linear factors, repeated linear factors, improper fractions), integration by partial fractions. ⇒ Integration by parts (examination questions will involve no more than two applications).	O1	Integrate standard functions such as $\cos 5x$, e^{ax} , $\frac{1}{ax}$, $\frac{f'(x)}{f(x)}$.
	O2	Clarify what is meant by the term “partial fraction”.
	O3	Find partial fractions for linear and repeated linear factors.
	O4	Find partial fractions for improper fractions.
	O5	Use substitution (including trigonometric forms) to evaluate indefinite and definite integrals (other than in simple cases, the substitution will normally be given). Trigonometric substitutions will involve sine and cosine only.
	O6	Carry out integration using partial fractions.
	O7	Carry out integration by parts.

APPENDIX A - EXEMPLAR TEACHING PLAN

Please note that the table below allocates a minimum of 4 hours contact time per week. Study Centres may need to schedule additional time depending on the ability of each cohort.

Method of delivery notation:

L – Lectures

E – Exercises

T – Tutorials

O – Other (specified)

Week	Hours	Topic	Teacher Guidance/Suggested Activities
1	1	A - Linear Equations	Identify the gradient and intercept in the equation of a straight line; calculate gradients of the normal. Find the distance between two points and the co-ordinates of the midpoint. (L/E) There may be considerable variation in the algebraic ability of students. Any practice in this area will probably be good for independent and directed study.
	1	A - Linear Equations	Find equations of other lines which are parallel to or perpendicular to the equation of a particular line. Solve pairs of linear simultaneous equations using elimination, substitution and graphical methods. (L/E)
	1	A - Linear Equations	Continue with simultaneous equations. Carry out practice examples on linear equations. (Mostly E)
	1	B - Simple Probability	Evaluate the probability of a single event. Familiarity with the notation $P(A)$ and $P(A')$ and realise that $P(A') = 1 - P(A)$. (L/E/T)
2	1	B - Simple Probability	For two events A and B , work out the probability of both happening, and of either happening. (Set notation is not required at this level – neither will candidates be expected to be aware of mutual exclusivity). Construct a tree diagram and use it to work out combined probabilities. (Mostly L)
	1	C - Quadratic Functions and Equations	Carry out a completing the square process and be able to sketch the graph of a quadratic function. (L/E) Handling algebra is often a weak spot, so what was said about independent and directed study in the previous week probably applies here.
	1	C - Quadratic Functions and Equations	Solve quadratic equations by factorising, completing the square and using the quadratic formula. Candidates must be able to present answers in surd form and understand

			the significance of the discriminant (but will not be expected to evaluate the size of coefficients in the original equation which give, for example, two real roots). (L/E) If time runs short, there is scope for continuing the process of solving by factorisation in the following week.
	1	C - Quadratic Functions and Equations	Continuation of previous session. (T & mostly E)
3	1	C - Further simultaneous equations	Solve by substitution two equations – one of which is quadratic and the other linear. (E & mostly L)
	1	C - Equation of a Circle	The form $(x - a)^2 + (y - b)^2 = r^2$. Find the radius and coordinates of the centre of a circle given its equation and vice versa. Know that the radius of a circle is perpendicular to a tangent. Link with previous lesson on solving equations e.g. find the points of intersection between a line and a circle.
	1	C - Inequalities	Solve linear and quadratic inequalities, either by using algebra or graphical methods. Recognise open and closed intervals and understand what an integer is. (L/E)
	1	C - Remainder Theorem	Use the Factor Theorem to show a given monomial is a factor of a polynomial and the Remainder Theorem to find its remainder upon division if it is not a factor. Candidates will be expected to divide a polynomial by a monomial or quadratic expression, and to be able to factorise an expression completely. (L/E)
4	1	C - Remainder Theorem	Continuation of previous session. (Mostly E)
	1	D - Indices	Know that $x^a \times x^b \equiv x^{a+b}$, $x^a \div x^b \equiv x^{a-b}$, $(x^a)^b = x^{ab}$, $x^{-n} \equiv \frac{1}{x^n}$ and $x^{a/b} \equiv \sqrt[b]{x^a}$. (L/E)
	1	D - Indices	Find the exact solutions of equations which use indices. (L & mostly E)
	1	D - Exponential Functions	Understand what an exponential is and be able to sketch the graphs of $y = e^x$, $y = e^{-x}$ and $y = a^x$. (Mostly L) Investigating exponential growth and decay could provide good material for independent or directed study.
5	1	D - Exponential Functions	Relate exponential change to a real situation e.g. population growth and decomposition of a solid into a liquid or a gas. (L/E/T)
	1	D - Logarithmic Functions	Understand how a logarithm behaves and compare it to real-life situations such as the Richter and pH scales. (Mostly L)

	1	D - Logarithmic Functions	Establish the connection between logarithms and exponentials. (L/E) Practice at using and manipulating logs would be invaluable time spent for independent and directed study.
	1	D - Logarithmic Functions	Use the logarithmic laws and apply them to solve equations and simplify expressions. (L & mostly E)
6	1	D - Logarithmic Functions	Consolidate on previous work and practice the use of logarithms. (E/T)
	1	E - Binomial Expansions	Revise expansion of quadratic functions. Use Pascal's triangle to find binomial coefficients and be able to expand completely $(1+x)^n$ and $(a+b)^n$ where n is a small positive integer (normally not more than 5). (L/E)
	1	E - Binomial Expansions	Continuation of previous session. (Mostly E)
	1	E - Binomial Expansions	Consolidate the previous two sessions. Candidates will be expected to pick out single terms in the expansion of $(a+b)^n$ where n is any integer. Candidates must also know what a coefficient is. (L/E/T)
7	1	E - Progressions	Start on progressions and introduce the difference between an Arithmetic Progression (AP) and a Geometric Progression (GP). Identify the first term and common difference in an AP. (E & mostly L)
	1	E - Progressions	Use the relevant formulae to find the nth term and the sum of the first n terms of an AP and a GP. (L/E) Candidates could be asked to find the first term of a GP, or how many terms are needed for geometric series, to exceed a certain value. This should only be done once logs have been covered.
	1	E - Progressions	Candidates should be able to find the common difference, common ratio and first term having been given the n th term or sum of the first n terms. (L & mostly E)
	1	E - Progressions	Use sigma notation and be able to generate a progression from a sigma expression. Generate the first few terms of a simple recurrence relation. Understand the idea of convergence and the conditions needed. Find the sum to infinity of a convergent series. (L/E)
8	1	F - Trigonometric Functions	Understand the definition of a radian; convert angles from degrees to radians and vice versa. Calculate the sin, cos and tan of any angle and be familiar with their graphs. (L/E)

	1	F - Trigonometric Functions	Know how to find the inverse of a trigonometric expression and solve simple trigonometric equations. Candidates will also need to be able to quote and recognise the exact values of the trigonometric functions of 0, 30, 45, 60 and 90 degrees and their radian equivalents. (L/E)
	1	F - Trigonometric Functions	Consolidate the previous two sessions and practice at examples – particularly solving the equations and identifying all the angles in each interval. (E/T) It is probably a case of the more time that students spend at practising examples in independent and directed study, the better.
	1	F - Trigonometric Functions	Apply the sine and cosine formulae to a non-right-angled triangle and use the formula for the area of a triangle $\frac{1}{2}ab \sin C$. Candidates should be familiar with the identity $\cos^2 x + \sin^2 x \equiv 1$ and with $\tan x = \frac{\sin x}{\cos x}$. (L/T)
9	1	G - Differentiation	Revision of finding the gradient of a straight line; understand the definition of the gradient of a curve at a point and, using first principles, find an expression for this gradient (using single integral powers of x only). (Mostly L)
	1	G - Differentiation	Using first principles, find gradient functions of other small powers of x . Introduce the idea of differentiating any power of x . (Mostly L)
	1	G - Differentiation	Use $\frac{dy}{dx}$ and $f'(x)$ and be able to differentiate any polynomial and any power of x . (L/E)
	1	G - Differentiation	Practice of differentiating polynomials and substituting in values of x . (Mostly E)
10	1	G - Differentiation	Extend differentiation to find the equations of a tangent and a normal to a curve at a specified point. (L/E)
	1	G - Differentiation	Extend the process to differentiation of exponentials (expressions of the form $y = e^{kx}$) and logarithms. (L/E)
	1	G - Differentiation	Differentiation of trigonometric expressions ($\sin x$, $\cos x$ and $\tan x$ only). Use of the formula booklet to obtain derivatives of standard functions. (L/E)
	1	G - Differentiation	Consolidate on the previous two weeks' work with plenty of practice at differentiation. (E/T) Plenty of scope here for practice in independent and directed study.

11	1	G - Differentiation	Find the second derivative of a function and be familiar with the notation $\frac{d^2y}{dx^2}$ and $f''(x)$. (L/E)
	1	G - Differentiation	Find the stationary points of a function. (L/E) Points of inflexion will be identified only in instances of zero gradient.
	1	G - Differentiation	Determine the nature of any stationary point. (L/E)
	1	G - Differentiation	Sketch the graph of a function once the turning points are known. Sketch the graph of a derivative. (Mostly E)
12	1	G - Differentiation	Extend the process of finding stationary values to practical optimisation problems. Typical cases could be to find the maximum volume of a solid which has a fixed surface area, and to find the minimum surface area of a solid with a fixed volume. (Mostly L) Other examples could be maximising an area of a shape with fixed perimeter.
	1	G - Differentiation	Extension of previous session. (L/E)
	1	G - Differentiation	More practice at optimisation problems and start of a general consolidation of differentiation. (Mostly E) In view of the size of this topic, a long consolidation will probably be necessary.
	1	G - Differentiation	Completion of the consolidation started the previous session. (E/T)
13	1	H - Integration	Identify integration as the inverse of differentiation. Integrate any power of x . (Mostly L)
	1	H - Integration	Integrate e^{kx} where k is a constant; integrate $\sin x$ and $\cos x$. (L/E)
	1	H - Integration	Realise the integration of $\frac{1}{x} = \ln x + c$. Use the formula booklet to determine indefinite integrals including integrals of the form $\frac{1}{ax+b}$ where a and b are constants. (L/E) Practice at integration would probably be time usefully spent in independent and directed study.
	1	H - Integration	Find the indefinite integral of any of the functions above. (Mostly E)
14	1	H - Integration	Evaluate definite integrals. (L/E)

	1	H - Integration	Apply definite integrals to finding the area between the curve and the x – axis. (L/E)
	1	H - Integration	Find more difficult areas, including those below the x –axis. (L & mostly E) Students should realise that areas below the x – axis will give negative values and appreciate the meaning of an integral giving zero.
	1	H - Integration	Extend integration to finding an area which is bound by two or more curves. (E/T)
15	End of Semester 1 Test		
16	1	I - Introduction to Statistics	Distinguish between, and give examples of, continuous and discrete data. Evaluate mean, mode, median and range. (L/E)
	1	I - Introduction to Statistics	Calculate a standard deviation and understand what it means. (L/E)
	1	I - Introduction to Statistics	Use grouped frequencies to estimate the mean and standard deviation. (L/E) Understand why these are only estimates.
	1	I - Introduction to Statistics	Draw a line graph to illustrate discrete data and a histogram to illustrate continuous data. (L/E) Students should take care to label axes and will be expected to scale them sensibly.
17	1	I - Introduction to Statistics	Evaluate cumulative frequency and construct a cumulative frequency graph. (L/E) Students should take care to label axes and will be expected to scale them sensibly.
	1	I - Introduction to Statistics	Use the graph to estimate the median, quartiles and interquartile range. Identify possible skew in a distribution, giving a reason. (L & mostly E) Students should take care to label axes and will be expected to scale them sensibly.
	1	J - Further Probability and Set Theory	Be familiar with set notation (knowledge of the following symbols will be expected: $\cap$, $\cup$, $\emptyset$, $\in$ and $\prime$). (L/E)
	1	J - Further Probability and Set Theory	Construct, and interpret, Venn diagrams. Use them to find probabilities. (Mostly E)
18	1	J - Further Probability and Set Theory	Construct, and interpret, more complicated tree diagrams. (L/E)

			Tree diagrams will not have more than three branches from any given point.
	1	J - Further Probability and Set Theory	Combine probabilities using tree diagrams and Venn diagrams. (L/E)
	1	J - Further Probability and Set Theory	Continuation of previous session (mostly E)
	1	J - Further Probability and Set Theory	Distinguish between independent and mutually exclusive events. Calculate conditional probabilities. (L/E/T) Candidates will need to be aware of $P(A \cap B) = 0$ for mutually exclusive events and $P(A \cap B) = P(A) \times P(B)$ for independent events.
19	1	K - Correlation	Draw a scatter graph and the line of best fit. (L/E) Candidates will not have to carry out significance tests.
	1	K - Correlation	Obtain some idea of sign and strength of correlation. (L/E)
	1	K - Correlation	Calculate the product moment correlation coefficient and relate its value to the strength and type of correlation, with particular reference to values close to -1, 0 and 1. Appreciation that correlation does not imply causation. (L & mostly E)
	1	K - Correlation	Continuation of previous session and further practice at calculating and interpreting correlation. (Mostly E)
20	1	K – Linear Regression	Compute the equation of a least-squares linear regression line for random variable Y and non-random variable X . (L/E) The above also applies concerning the numbers of pairs of readings. Students may have to draw the equation on a scatter graph.
	1	K – Linear Regression	Use the equation to estimate values but be aware of the reliability of estimates. The equation will normally be in the form $y = ax + b$ and candidates will be expected to know what a and b represent, and to recognise if their values make sense. (L/E) Questions asking for an equation of X on Y in the form $x = ay + b$ will not be set.
	1	K - Time Series	Calculate the moving averages of a time series. (L/E)

	1	K - Time Series	Construct a time series chart and draw a trend-line. Candidates may be asked to extrapolate the trend line for a short distance but may also be asked to explain why extrapolation cannot be relied on. (Calculating seasonal variations and residuals will not be expected.) (L/E) Any number of points could be asked, but they will normally be 3 or 4 point.
21	1	L – Probability Distributions	Understand what a distribution is and be able to recognise and describe one. Distinguish between discrete and continuous distributions. (Mostly L)
	1	L – Probability Distributions	Construct a probability distribution relating to a given situation involving a discrete random variable X . Evaluate the expected value of a random variable. Appreciate the connection between expected value and mean. (L/E)
	1	L – Probability Distributions	Evaluate the variance and standard deviation. Find the expected value and variance for expressions like $3X - 2$. (L/E) Students can usefully use independent and directed study time practising typical examination questions on this topic.
	1	L – Probability Distributions	Describe the binomial distribution and perform simple calculations involving small values of n . (L/E)
22	1	L – Probability Distributions	Perform calculations using the binomial distribution and larger values of n . Use the cumulative binomial distribution tables. Appreciate important assumptions when using the binomial distribution. (L/E)
	1	L – Probability Distributions	Describe the Normal distribution. Convert data into standardised form. Be aware of the limitations of the Normal distribution (e.g. the assumption that there is no restriction on the values taken by x but this is not usually the case in reality). (Mostly L and some E) Students can usefully spend time in independent and directed study practising examples in the binomial and Normal distributions.
	1	L – Probability Distributions	Continuation of previous session. (Mostly E)
	1	L – Probability Distributions	Set up a confidence interval (any samples will be taken from a background distribution which is assumed to be Normal with a known standard deviation). Explain what is meant by a confidence interval. (L/E/T) Students should be aware of the Central Limit Theorem but will not have to apply it in an examination.

23	1	M – Financial Mathematics	Be thoroughly familiar with all types of percentage. Calculate any direct and reverse percentage and evaluate any percentage change and error. (L/E/T)
	1	M – Financial Mathematics	Identify what type of percentage is required in a miscellany of examples. (L/E/T)
	1	M – Financial Mathematics	Appreciate what interest on an investment is. Calculate simple interest and compound interest. (L/T) Compare two types of investment to see which gives the better interest. (Mostly E) Students can use the compound interest formula and may need to draw on their knowledge of logs.
	1	M – Financial Mathematics	Consolidate on previous week and extend the ideas to finding, for example, the total interest as a percentage of what was originally invested. (E/T)
24	1	M – Financial Mathematics	Calculate appreciation and depreciation. (L/E) In this session and the next, students will be expected to work out the value of an article after any number of years by applying the idea of a geometric progression (specific learning outcome E8). They will also be expected to work out the expected time that a certain value is reached by applying logarithms (specific learning outcome D6). (L/E) Examples may be the price of property or a car.
	1	M – Financial Mathematics	Consolidate on appreciation and depreciation. (E/T) There must be an awareness of why mathematical models may not be reliable in the long run (e.g. interest rates may change).
	1	N – Further Differentiation	Know that the differentiation of a^x is $a^x \ln a$. Apply the product rule. (L/E)
	1	N – Further Differentiation	Apply the quotient rule. (L/E) Candidates may be asked to derive certain results (e.g. showing that the derivative of $\tan x$ is $\frac{1}{\cos^2 x}$).
25	1	N – Further Differentiation	Apply the chain rule. (L/E)
	1	N – Further Differentiation	Practise examples using a mixture of the above rules. (L/T) Practice at differentiation in independent and directed study time will be invaluable.
	1	N – Further Differentiation	Understand what an implicit function is and how to apply differentiation. (Mostly L)

			In this module, candidates will be expected to draw on concepts learnt in Topic G.
	1	N – Further Differentiation	Find $\frac{dy}{dx}$ of an implicit function which could involve use of the product rule. (L/E)
26	1	N – Further Differentiation	Find the equations of a tangent and a normal at a point. (L/E)
	1	N – Further Differentiation	Find stationary values. (L/E) Again, time spent practising in independent and directed study will be invaluable.
	1	O – Further Integration	Integrate standard functions. Recognise the integral of $\frac{f'(x)}{f(x)}$. (L/E) e.g. $\cos 5x$, e^{ax} , $\frac{1}{ax}$.
	1	O – Further Integration	Continuation of previous session. (Mostly E)
27	1	O – Further Integration	Resolve expressions into partial fractions. (L/E) Expressions will have linear and repeated factors only, with no quadratic factors.
	1	O – Further Integration	Resolve improper fractions into partial fractions. (L/E)
	1	O – Further Integration	Integrate expressions which have been resolved into partial fractions. (L/E)
	1	O – Further Integration	Use substitution to evaluate indefinite and definite integrals. (L/E) Other than in simple cases, the substitution will normally be given.
28	1	O – Further Integration	Continuation of previous session. (Mostly E)
	1	O – Further Integration	Use integration by parts. (L/E) Integration by parts will need to be applied not more than twice.
	1	O – Further Integration	Consolidation of integration by substitution and by parts. (Mostly E)
	1	O – Further Integration	Practice at recognising and performing any integral. (E/T)

			Regarding independent and directed study, the more practice the better.
29	4	Revision	
30	Exam Week		

APPENDIX B – NOTATION LIST

The notation below may be used in IFY Maths Business assessments without explanation. If forms of notation other than those listed are used in an assessment they will be defined within the question.

∞	Infinity
a	First term for an arithmetic or geometric sequence
d	Common difference for an arithmetic sequence
r	Common ratio for a geometric sequence
S_n	Sum to n terms of a sequence
S_∞	Sum to infinity of a sequence
$\sum_{i=1}^n a_n$	$a_1 + a_2 + \dots + a_n$
$\sqrt{a}$	the non-negative square root of a
$ a $	the modulus of a
$\frac{dy}{dx}$	the derivative of y with respect to x
$f'(x)$	the derivative of $f(x)$ with respect to x
$\frac{d^2y}{dx^2}$	the second derivative of y with respect to x
$f''(x)$	the second derivative of $f(x)$ with respect to x
$\int y \, dx$	the indefinite integral of y with respect to x
$\int_a^b y \, dx$	the definite integral of y with respect to x between the limits $x = a$ and $x = b$
$\binom{n}{r}, {}^nC_r$	the binomial coefficient
$\sin, \cos, \tan$	the trigonometric functions
$\sin^{-1}, \cos^{-1}, \tan^{-1}$	the inverse trigonometric functions
$A \cup B$	union of the events A and B
$A \cap B$	intersection of the events A and B
$P(A)$	probability of the event A
A'	complement of the event A
$P(A B)$	probability of the event A given the event B
$E(X)$	expectation of the random variable X
$Var(X)$	variance of the random variable X
$\bar{x}$	sample mean
s^2	sample variance
s	sample standard deviation

μ	population mean
σ^2	population variance
σ	population standard deviation
z	standardised normal random variable
H_0	null hypothesis
H_1	alternative hypothesis
S_{xx}, S_{yy}, S_{xy}	summary statistics

APPENDIX C - FORMULAE BOOKLET

Refer to separate Formulae Booklet.

It is the centre's responsibility to print "clean" (new) copies of the Formula Booklet for the End of Semester 1 Test and the Final Examination.

Syllabus Update History: NCUK International Foundation Year Mathematics (Business)

Release Date	Version No.	Summary of Main Changes	Author
June 2023	4.3	Updates for 2023/24	Product Team
May 2022	4.2	Updated for 2022/23	Product Team
Sept 2021	4.1	Link to electronic version of textbook updated after change by publisher	Product Team
June 2021	4.0	Full Review	Debby Gill, Peter Davies & Product Team

This syllabus is valid for the 2023-24 academic year only.

NCUK – The University Consortium

WeWork, No1 Spinningfields, Quay Street, Manchester, M3 3JE, United Kingdom

Tel: +44 (0)161 549 9220

www.ncuk.ac.uk

©NCUK Ltd.

NCUK
THE UNIVERSITY CONSORTIUM